DARM: Mathematical Correction Report

Proof Audit, Notation Repairs and Preprint-Release Assessment

Callum Connell

DARM Preprint v1.0
19 October 2026

Abstract

Independent audit of the DARM manuscript through Experiment 18, documenting substantive mathematical, notation and reporting changes, the scope of verified proofs, and remaining prerequisites for a credible public preprint. The audit is of the supplied manuscript and source archive, not an independent rerun of historical financial or original Monte Carlo data-generating code.

1 Substantive corrections

1. **Theorem 5.1: double subtraction of the contemporaneous martingale shock (v1.0 correction).** The previous proof stated incorrectly that

$$h_t - \mathbb{E}h_t = \frac{1}{m}\sum_{j\geq 1}\psi_j\zeta_{t-j} - \frac{\zeta_t}{m}.$$

With $u_t = A_t h_t$, $m = \mathbb{E}A_t$, $\zeta_t = u_t - mh_t$ and $Y_t = u_t - \mathbb{E}u_t$, the correct renewal representation is $Y_t = \sum_{j\geq 0}\psi_j\zeta_{t-j}$ with $\psi_0 = 1$; simultaneously $Y_t = \zeta_t + m(h_t - \mathbb{E}h_t)$. Therefore

$$\boxed{h_t - \mathbb{E}h_t = \frac{Y_t - \zeta_t}{m} = \frac{1}{m}\left(\sum_{j\geq 0}\psi_j\zeta_{t-j} - \zeta_t\right) = \frac{1}{m}\sum_{j\geq 1}\psi_j\zeta_{t-j}.}$$

After shifting the summation start to $j = 1$, subtracting ζ_t/m again is algebraically false and would spuriously introduce a lag-zero term into a predictable conditional variance. The corrected strict-causal filter is $D_m(s) = \rho W(s)/[1 - \rho m W(s)]$, and $\varepsilon_t^2 - \mathbb{E}\varepsilon_t^2 = \chi_t + D_m(L)\zeta_t$ for $\chi_t = \varepsilon_t^2 - h_t$. The corrected formula confirms the original absolute covariance summability conclusion and preserves the $m = 1$ main-text spectral formulas, including the contemporaneous $\mathrm{Cov}(\chi_t, \zeta_t)$ cross term. The technical supplement now states and derives the general-m spectra and qualifications. Neither the data-based results nor the theorem's substantive conclusion changes. This is a *targeted proof correction*, not certification that every theorem has undergone an independent formal proof verification.

2. **Martingale innovation assumption.** The incorrect assertion $\mathbb{E}[z_s \mid \mathcal{F}_{s-1}] = 1$ is replaced by $\mathbb{E}[z_s \mid \mathcal{F}_{s-1}] = 0$ and $\mathbb{E}[z_s^2 \mid \mathcal{F}_{s-1}] = 1$. For h_s nonnegative, integrable and $\mathcal{F}_{s-1}$-measurable, $d_s = h_s(z_s^2 - 1)$ is an integrable martingale difference. By the tower property, $\mathbb{E}_t RV_{t,H} = \mathbb{E}_t IV_{t,H}$. The zero conditional first moment is needed for martingale-difference *returns* and cumulative-return conditional-variance interpretation, not for equality of these two quadratic-variation means. An ARCH(1) counterexample shows that $RV - IV$ need not be independent of future IV.

3. **Exact asymmetric GJR parameter domain.** The geometric mapping is $\alpha_G = \rho(1-\beta)(1-\kappa)$, $\gamma_G = 2\rho(1-\beta)\kappa$, $\omega_G = (1-\beta)\omega_D$. With $0 \le \kappa < 1$, a nonzero feedback process maps to $\alpha_G > 0, \gamma_G \ge 0$ and $\alpha_G + \gamma_G/2 + \beta < 1$. The boundary $\alpha_G = 0, \gamma_G > 0$ would require $\kappa = 1$, outside the maintained domain. At $\rho = 0$, $\alpha_G = \gamma_G = 0$ and κ is not identified. Negative γ_G is not represented.

4. **Transformed-shock notation.** The geometric recursion telescopes with $u_t = A_t h_t$; ν is exclusively the Student-t degrees-of-freedom parameter. The erroneous use of ν as the substituted shock is corrected.

5. **One-lag fourth-moment identity.** For nonsymmetric innovations, the previous numerator $1+\rho$ and factor $1-\rho$ are valid only if $m = \mathbb{E}A = 1$. The corrected expression is

$$\mathbb{E}h_t^2 = \frac{\omega^2(1+\rho m)}{(1-\rho m)(1-\rho^2\mathbb{E}A^2)}$$

   provided $\rho^2\mathbb{E}A^2 < 1$ and the causal stationary positive-intercept solution is used.

6. **White-noise notation.** The transformed-shock martingale innovation is uniformly denoted $\zeta_t = u_t - mh_t$ and the squared-return martingale innovation $\chi_t = \varepsilon_t^2 - h_t$. For symmetric innovations $m = 1$, allowing $u - Eu = [1 - C(L)]^{-1}\zeta$ and $\varepsilon^2 - E\varepsilon^2 = \chi + D(L)\zeta$. The contemporaneous covariance between ζ_t and χ_t must appear in the squared-return spectrum.

7. **No universal long memory.** Under finite fourth moments and subcritical ARCH feedback, the renewable coefficients are summable and so are squared-return covariances. A finite power-law lag kernel need not generate long memory in the econometric non-summability sense. Near-unit feedback and slow pre-asymptotic decay remain possible and are not ruled out by this statement.

8. **Statistical inference.** Coverage of likelihood-ratio intervals under interior true parameters does not become a boundary-mixture theorem merely because a numerical optimiser returns box-edge solutions. Experiment 15 exploratory bootstrap samples are too small to establish superiority of bootstrap correction.

9. **Forecast targets and losses.** Expected oracle-relative QLIKE regret is a conditional-mean scoring functional, while realised paired QLIKE differences use matched paths; they are not interchangeable. The existence of $\mathbb{E}_t RV = \mathbb{E}_t IV$ does not validate IV predictive intervals.

2 Theorem-by-theorem verification status

Result	Audit finding and exact scope
Causal existence and uniqueness	Volterra path expansion is nonnegative; under i.i.d. A_t, normalised nonnegative weights and $\rho m < 1$, Tonelli yields a finite-mean causal solution. Uniqueness is in the class of integrable causal stationary solutions on the same noise; no universal nonintegrable uniqueness is asserted. Ergodicity follows from Bernoulli-shift measurability.
Second and higher moments	$\mathbb{E}h = \omega/(1-\rho m)$ under integrability. The L^q condition $\rho\|A\|_q < 1$ is sufficient, not sharp at general K. At $K = 1$, $\rho^2\mathbb{E}A^2 < 1$ is the exact positive-intercept finite-second-moment threshold. For return fourth moments additionally require $\mathbb{E}z^4 < \infty$.

Result	Audit finding and exact scope
Infinite geometric GARCH	Telescoping and vanishing L^1 remainder are valid for $\beta < 1$ with integrable stationary h; the finite truncation is not algebraically GARCH(1,1).
Asymmetric geometric GJR	Corrected closed/open boundary restrictions above. The $\kappa = 1$ excluded boundary and $\rho = 0$ identification failure are both explicit.
Power-law weight identification	Adjacent weight ratios identify δ, then α, when $K \geq 3$, $\alpha, \delta > 0$. For $\alpha = 0$ or $K \leq 2$ the two-parameter map is not identified by the weight vector. This does not imply identification from the return law.
Monotonicity of kernel summaries	Under finite K, positive weights, and interior positive shape coordinates, derivative signs follow covariance-order arguments. Effective count is a concentration statistic, not the response half-life.
Renewal coefficients and dominant pole	Under finite K, $0 < \rho < 1$, $w_1 > 0$, $C(R) = 1$ has a unique $R > 1$; $s = R$ is the unique zero within $\lvert s\rvert \leq R$, and $\psi_j \sim R^{-j}/[RC'(R)]$. In periodic supports without $w_1 > 0$, other equally dominant poles can invalidate a single-term equivalent.
Spectral representation	Conditional-variance spectrum and squared-return spectrum differ for $\kappa > 0$. The squared-return spectrum includes the $2\,\mathrm{Cov}(\zeta, \chi)\Re D$ cross term. Fourth moments and stationarity are required.
Random companion moments	$\mathrm{spr}\{\mathbb{E}(M \otimes M)\} < 1$ with $\mathbb{E}A^2 < \infty$ and a finite-mean causal stationary state is sufficient for a state second moment. General necessity is *not* asserted without nondegeneracy. Negative Lyapunov exponent, with logarithmic integrability, is a separate sufficient strict-stationarity route for finite-dimensional affine recursions.
Multistep forecasts	The deterministic state companion gives $\mathbb{E}_t h_{t+j}$, with future A replaced by $m = \mathbb{E}A$; geometric convergence holds when feedback is subcritical and the finite positive kernel is primitive. Future shocks are not substituted with realised unseen values.
Kernel Gram matrix	Positive definite for $K \geq 3$ and interior parameters; a small eigenvalue explains geometric near-equivalence, but it is not itself the fitted likelihood information matrix after nuisance profiling.
Expected versus realised QLIKE	Regret $\log(v/v^\star) + v^\star/v - 1$ is nonnegative for fixed conditional oracle means. Realised paired losses can exhibit distinct finite-sample rankings; the paired test is over independent datasets, not forecast origins.

3 Mathematical limitations remaining

The infinite-order Volterra construction is proved in the integrable causal class; a full classification of nonintegrable strict stationary solutions is outside the paper. General finite-K necessary-and-sufficient moment thresholds are not claimed. The finite-pole formula concerns responses to transformed martingale shocks and must not be confused with an economically structural shock-sign impulse response. Near-boundary Wilks behaviour and all return-law identification results need separate regularity and model-specific likelihood arguments.

4 Scientific and reproducibility assessment

The historical independent ten-day comparison remains $\Delta\mathrm{QLIKE} = 0.015048$, HAC $p = 0.146291$, Holm-adjusted $p = 0.292582$; it does not establish superiority. Synthetic Experiments 13–18 illustrate the coexistence of weak structural estimation and useful point forecasts, plus misspecification-

dependent latent-risk interval failure. No causal experiment proves why different log-SV regimes diverge: the regimes jointly vary parameters. The new Experiments 13–18 archive includes authored source implementations and notebooks for all six experiments, as well as reported tables, execution outputs in 13–15, replication-level score records in 16–18, and validation scripts. It still lacks many original paths, optimiser checkpoints and an exact execution-environment lock. This is executable source availability, not certification of complete end-to-end reproduction.

5 Preprint-release conditions

Before journal submission or strong claims of reproduction, release a documented and legally shareable point-in-time market-data construction, precise constituent-selection rules, environment/seed manifests, fitted-parameter checkpoints, forecast-origin loss arrays, original simulation/oracle paths and the full interval-draw procedure, and validate source-to-output reproducibility under pinned dependencies. Date and publicly archive the exact final manuscript and research code; distinguish internal "frozen script" documentation from prospective preregistration. Add independent proof review of any extension beyond the restricted theorem domains stated above. Related ARCH(∞) dependence, strict-stationarity and likelihood-limit results are established in Giraitis et al. (2000), Bougerol and Picard (1992) and Robinson and Zaffaroni (2006). No extra Monte Carlo results are claimed by this editorial audit.

References

Bougerol, P. and N. Picard (1992). "Stationarity of GARCH Processes and of Some Nonnegative Time Series". In: *Journal of Econometrics* 52.1–2, pp. 115–127. DOI: 10.1016/0304-4076(92)90067-2.

Giraitis, L., P. Kokoszka, and R. Leipus (2000). "Stationary ARCH Models: Dependence Structure and Central Limit Theorem". In: *Econometric Theory* 16.1, pp. 3–22. DOI: 10.1017/S0266466600161018.

Robinson, P. M. and P. Zaffaroni (2006). "Pseudo-Maximum Likelihood Estimation of ARCH(∞) Models". In: *The Annals of Statistics* 34.3, pp. 1049–1074. DOI: 10.1214/009053606000000245.